Data Engineering 300 Final Project Proposal
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Introduction

Music streaming platforms generate massive volumes of rapidly changing data, including listening behavior, artist metadata, and song popularity. Designing a system that can efficiently ingest, process, and analyze this data at scale can help inform decisions such as marketing budgets and concert demand, and techniques for building scalable pipelines in this domain can be applied similarly to other fields.

We are interested in designing a reusable music analytics pipeline that efficiently ingests raw music from streaming platforms and converts it into a unified, analysis-ready format to gain insights and make predictions about trending songs, artists, and genres over time. We plan to combine real-time API ingestion with historical datasets and utilize distributed computing techniques to clean, normalize, and store streaming data in a form suitable for analysis. The system will allow us to explore trending songs, artists, genres, and investigate early signals associated with rising popularity.

Data Sources

Our primary data source will be the Spotify Web API, which provides access to artist metadata, track popularity metrics, audio features, and release information. We plan to retrieve this data using authenticated API requests and periodically refresh it to capture changes in popularity over time. Data ingestion will be designed to support incremental updates (instead of full reprocessing) and will be performed in batches due to rate limits imposed by the API. Other data sources include publicly available Last.fm datasets with large-scale user listening histories and timestamps. Using these bulk datasets, we can analyze long-term trends and temporal patterns that would be impractical to capture fully with the Spotify API alone. We expect to work with datasets ranging in several gigabytes, depending on the ingestion scope and the time window analyzed.

Goal Definitions

We are interested in achieving the following goals for this project:

1. Develop an automated data pipeline that ingests raw data from multiple sources and transforms it into an analysis-ready dataset. After performing data cleaning tasks using methods covered in class, such as data imputation and schema normalization across sources, we will then store our cleaned data in a **relational database** (such as Postgresql) to support querying and exploratory analysis.
2. Be able to perform analysis on our dataset to identify trends in song/artist popularity over time. As we are covering a very large domain with large amounts of data, we intend to use **PySpark** to distribute the computations used in aggregating and analyzing our data. We will then explore whether early indicators, such as a rapid rise in listen count, release timing, artist popularity trajectory, or genre-specific behavior, can predict which songs or artists are likely to trend in the near future. We will not be able to make long-term predictions given the industry's volatility, nor will we be able to draw conclusions about social causes of these trends, such as social media posts, cultural significance, or advertising campaigns.

Data Source References

- Spotify Web API. https://developer.spotify.com/documentation/web-api/
- Last.fm Dataset. https://www.last.fm/api